



Capstone Project Phase B

Evaluating Explainable AI Methods in Domains of Varying Human Impact

[GitHub](#)

Project Number

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1 Introduction

In recent years, AI's role in decision-making has evolved from rule-based systems that automated simple tasks to advanced machine learning models capable of supporting complex and strategic decisions [13].

While AI is widely integrated across various domains and plays an important role in critical decision-making [3], its impact varies significantly by risk level, presenting unique challenges and opportunities [13].

A central challenge is the increasing complexity of AI models, which has led to the 'black box' problem, where internal decision-making processes are hidden from users, making trust and reliability difficult to assess [22]. This opacity presents a significant challenge for user trust, which varies across domains. While low-risk scenarios often rely on emotional trust based on intuition, high-risk domains demand cognitive trust, which depends on understanding the system's accuracy and reliability [1].

Establishing trust between users and AI-based systems therefore requires explaining both the decision-making process and its outcomes. To address this trust gap, the field of Explainable Artificial Intelligence (XAI) has emerged to improve transparency and interpretability in complex AI systems. XAI provides tools that help users understand how and why a model produces a specific prediction [11]. It offers a range of explanatory approaches that vary in their level of detail and informativeness, including feature importance methods, visualization techniques, and counterfactual explanations. Additionally, explanatory depth is supported by both local explanations, which focus on individual predictions, and global explanations, which provide insights into the overall behavior of the model [16].

Therefore, this research included two experiments. In the first experiment, we examined how users perceive and trust AI systems across domains with different risk levels by asking participants to rate their level of trust in each domain. In the second experiment, we investigated user preferences toward different XAI explanation methods by evaluating the explanations using the SCS and S-TIAS frameworks, focusing on factors such as clarity, understanding, and trust in the AI system.

2 Background and Related Work

2.1 AI in Decision Making

Today, AI is widely integrated across various domains and plays an important role in supporting human decision-making. For instance, in the financial domain, AI decision making systems have been integrated into areas such as credit scoring, fraud detection, and investment management by analyzing historical data and assessing risk more efficiently. Similarly, in healthcare, AI assists in disease diagnosis, treatment planning, and resource allocation, while reducing human error and improving patient outcomes through the analysis of large medical datasets and medical images. Another example is in the logistics sector, which relies on AI to optimize supply chains, improve route planning, manage inventory, and enhance customer service through real-time data analysis, leading to improved operational efficiency and reduced costs [3].

However, despite these advantages, the integration of AI into decision-making presents significant challenges. A primary challenge is data quality and availability, since reliable models depend on high-quality input, many organizations struggle with incompatible data, inconsistent formats, and a lack of data standards and supervision. Furthermore, integration with existing systems is often complex and resource-intensive, requiring both technological compatibility and time for employees to learn and adjust [13]. Beyond these technical challenges, the integration of AI presents ethical concerns, particularly regarding bias and fairness. Since AI systems rely heavily on historical data, they can reinforce and even amplify existing inequalities found in the datasets. This bias can arise from various sources, including the training data itself, algorithmic design, or implementation practices. Ultimately, these biases can affect the fairness, accountability, and transparency of the decision-making process [13].

2.2 AI Decision Making in High, Medium, and Low Risk Domains

AI's impact on decision-making varies across domains with different risk levels, each presenting unique challenges and opportunities [13]. High-risk domains are domains where decisions have a significant impact on our lives, such as medical decisions, where the consequences of bias, errors, or lack of transparency can have severe repercussions [2]. Medium-risk domains involve decisions that may not be life-threatening but significantly influence our lives, such as in the Human Resources (HR) field. In this domain, decision-making involves tasks related to recruitment, selection, and training, as well as retention decisions [8]. Finally, low-risk domains typically involve decision-making where the consequences of errors are minimal. For example, in the Text Analysis domain, the objective is to determine whether an input text was produced by a human or by an AI system [14] a task with limited direct impact on individuals' well-being.

To examine these differences, this section focuses on three distinct domains with different risk levels: Healthcare, Human Resources (HR), and AI Text Detection.

2.2.1 Healthcare (High Risk Domain):

AI is increasingly used in healthcare to support clinical decision-making and improve patient care. A common application is Clinical Decision Support Systems (CDSS), which analyze large volumes of medical data, such as, patient records, clinical guidelines, and scientific literature to assist healthcare professionals in diagnosis, treatment selection, and risk identification. In addition, AI and deep learning models are used to predict disease progression by analyzing historical clinical, laboratory, and imaging data, enabling more proactive and personalized treatment planning. Together, these applications help reduce human error and improve patient outcomes [3].

However, the use of AI in healthcare presents important challenges. A significant challenge is the lack of transparency, often referred to as the black-box nature of AI systems, which makes it difficult for clinicians to understand and interpret how decisions are generated. This can reduce clinicians' confidence in AI recommendations and limit their willingness to rely on such systems in clinical practice[2].

2.2.2 Human Resources (Medium Risk Domain):

Currently, AI-driven tools are increasingly used to support these decision-making areas across Human Resources. In recruitment and selection, AI applications automate resume screening and apply predictive hiring analytics to assess candidate suitability and forecast future performance. In employee training, AI enables adaptive learning platforms that tailor content to individual learning needs, while in performance management, it supports continuous feedback and data-driven evaluations. Additionally, AI applications contribute to retention strategies by monitoring employee engagement and predicting future workforce needs [9].

However, the integration of AI also presents several challenges and limitations. A central concern is algorithmic bias, as AI systems rely on historical data that may contain biases related to gender, race, or other factors. As a result, AI-driven tools may produce discriminatory outcomes [9]. Another significant limitation involves data privacy and security. AI-based HR systems depend on large volumes of sensitive employee data, raising concerns about unauthorized access, misuse, and compliance with data protection regulations. In addition, AI deployment in HR raises important ethical and legal considerations. The opaque nature of many AI algorithms can undermine trust and raise concerns regarding bias and discrimination, while organizations must also ensure compliance with labor laws and anti-discrimination regulations [9].

2.2.3 Text Analysis (Low Risk Domain):

The importance of these decisions increases as recent advances have improved the diversity and quality of machine-generated text, enabling the creation of human-like content at scale. These developments increase the difficulty of text detection and raise concerns regarding the misuse of generated text for purposes such as phishing, disinformation, fake reviews, academic dishonesty, and spam. Consequently, decision-making in AI-based text detection systems plays an important role in mitigating potential abuse and supporting the principles of trustworthy AI [5].

However, AI-based text detection faces several challenges. One major limitation is cross-domain performance, as detection models trained on one type of text often perform poorly when applied

to different domains. In addition, even minimal human editing of AI-generated text can significantly reduce detection accuracy, making reliable identification difficult. Ethical concerns also arise, including issues related to privacy, bias in detection systems, and unequal access to detection tools across languages and cultural contexts [12].

2.3 Human trust in AI system

The increasing use of AI systems in everyday applications highlights the importance of user trust in AI. Trust is a central component of the interaction between humans and AI systems, as it influences whether users accept and rely on AI-generated decisions. However, trust in AI changes depending on the decision's risk. In low-risk domains, users often rely on emotional trust, which is based on intuition or general feelings. In contrast, in high-risk domains, cognitive trust becomes essential and depends on the user's perception of the system's ability to make accurate and reliable decisions. As a result, trust in AI strongly depends on system-related factors such as accuracy, reliability, transparency, and explainability [1].

However, the need for transparency is directly challenged by the 'black box' nature of modern AI. While observers can witness the inputs and outputs of these systems, the complex and non-linear inner algorithms remain hidden. Consequently, the way AI reaches its conclusions is opaque. This opacity stems from two main causes. In some cases, algorithms are kept hidden simply to protect trade secrets. However, in Deep Learning, the problem is different: it arises from the system's complexity. Unlike simple code, these models use billions of parameters to make decisions. As a result, even if the code were fully visible, the internal logic would remain incomprehensible even to the experts who created it [22].

2.4 Explainable Artificial Intelligence (XAI)

Establishing trust between users and AI based systems requires explaining the decision-making process and its result. Explainable Artificial Intelligence (XAI) refers to a set of methods and techniques designed to make the decisions and outputs of AI systems transparent, interpretable, and understandable to users. As AI models become increasingly complex and are used in critical decision-making domains, many of these systems operate as “black boxes”. This lack of transparency limits users' ability to understand, trust, and effectively evaluate AI-driven decisions. XAI addresses the black-box problem by providing explanations that reveal how models arrive at specific conclusions, thereby enabling users to interpret, verify, and assess AI behavior. By improving transparency and interpretability, XAI supports trust, accountability, and ethical AI use, while also helping to identify potential biases and unintended consequences in high-stakes domains [16].

2.4.1 XAI Tools

The following section reviews three main XAI tools: LIME for local explanations, SHAP for global explanations, and DICE for counterfactual explanations.

2.4.1.1 LIME (Local Interpretable Model-Agnostic Explanations)

LIME is a widely used and flexible tool designed to interpret the predictions of complex "black box" systems. As a model-agnostic technique, it can be applied to any AI model because it does not need to see the model's internal parts, such as its layers or weights. Instead, LIME explains a specific decision by focusing on a local surrogate model [11].

It provides a local explanation that highlights specific features that led to a particular decision. While effective, the quality of the explanation depends on how well the simple model fits the complex one, and it can sometimes produce slightly different results for the same input because of the way it uses random sampling [11].

2.4.1.2 SHAP (Shapley Values)

SHAP is a tool that explains AI predictions using rules from Game Theory. It treats the prediction process like a game where all input features work together as a "team" to produce a result. SHAP calculates the credit each feature deserves, ensuring that if two features have the exact same effect, they receive equal credit [11].

It calculates how much each specific feature pushes the result away from this average, either increasing the score or decreasing it. The final explanation is simply the sum of these individual contributions, showing exactly how the combination of features led to the specific decision [11].

2.4.1.3 DiCE (Diverse Counterfactual Explanation)

DiCE is a XAI method used to provide insights into machine learning decisions by generating counterfactual explanations. The primary goal of this method is to identify the minimum changes needed in the input data to modify a model's specific prediction. By doing so, DiCE offers actionable insights, showing users exactly what would need to be different for the model to reach a desired outcome [21].

2.5 Summery

In conclusion, the literature demonstrates that AI systems are increasingly central to decision-making processes across diverse domains, each associated with different levels of risk and potential impact. Despite their growing adoption, these systems face significant challenges, most notably the opacity of black-box models, which limits user trust, accountability, and ethical oversight. To address these concerns, Explainable Artificial Intelligence (XAI) has emerged as a framework for enhancing transparency and interpretability, enabling users to better understand and evaluate AI-driven decisions. Furthermore, XAI techniques such as LIME, SHAP, and DiCE provide approaches to explaining model behavior through local explanations, feature attribution, and counterfactual reasoning. Collectively, the reviewed literature highlights explainability as an essential requirement for the trustworthy and responsible deployment of AI systems, particularly in domains where decision-making risk is high.

3 Research Process

In this project, we evaluated how different Explainable Artificial Intelligence (XAI) methods were perceived by users across domains with varying levels of risk. We focused on three domains: healthcare, human resources, and text analysis, representing high, medium, and low risk scenarios. The goal was to examine how users evaluated different explanation methods in terms of understanding, clarity, trust, and perceived usefulness, and whether these evaluations changed according to the domain's risk level.

The research began with an extensive literature review that helped us identify key challenges in AI-based decision-making and understand the role of explainability in improving user trust and understanding. Based on these insights, we selected three representative datasets, each corresponding to one of the selected domains.

Next, we trained three machine learning models, each on a different dataset, and generated predictions for further analysis. For each prediction, we applied three XAI methods: SHAP, LIME, and DiCE. The predictions and explanations were stored in a database and integrated into a web-based system that presented the AI decisions alongside their visual explanations.

After building the system, we designed and conducted the main part of the project: the user evaluation study. In this study, participants were exposed to different explanation types in a randomized manner and were asked to evaluate them using SCS (System Causability Scale) and S-TIAS questionnaires. These questionnaires measured aspects such as understanding, clarity, trust, and perceived usefulness of the explanations. The collected feedback was then analyzed to identify user preferences and examine whether explanation effectiveness changed across domains with different risk levels.

Finally, we analyzed the collected feedback to compare the explanation methods across the different risk domains and to examine how the perceived risk level of a domain influenced users' evaluations and preferences.

3.1 Architecture and tools

The following tools and technologies will be used in our project:

3.1.1 Backend (AI and Data Processing)

Programming Language:

- Python: used for data preprocessing, model training, predictions and explanations of the AI models.

Machine Learning Library:

- Scikit-learn: used to build and train all machine learning models, pipelines, and preprocessing steps

Machine Learning Models: The training environment was in Google Colab.

- Logistic Regression: used for the text analysis domain because it performs well on textual data and provides stable and consistent results for text classification tasks.
- Multi-Layer Perceptron (MLPClassifier): used for the healthcare and HR domains because these datasets involve complex and non-linear relationships between features, which neural networks are better suited to model.

Preprocessing Techniques:

- ColumnTransformer: used to apply different preprocessing to numerical and categorical features
- StandardScaler: used to normalize numerical features
- One-Hot Encoding: used to encode categorical features
- TF-IDF Vectorization: used to convert text into numerical vectors
- FeatureUnion: used to combine TF-IDF features with style features
- Style Feature Extraction from Text: used in the text analysis domain to generate 9 stylistic attributes, since DiCE explanations are difficult to apply directly on raw textual data. These features allow DiCE to produce meaningful explanations.
- MaxAbsScaler: used to scale style features

Model Training & Evaluation:

- Pipeline: used to combine preprocessing and model into a single workflow
- Stratified K-Fold Cross-Validation (k=5): used to evaluate the model.

XAI Tools:

- SHAP (SHapley Additive Explanations): Provides insights into individual predictions by calculating the contribution of each feature using principles from cooperative game theory.
- LIME (Local Interpretable Model-Agnostic Explanations): Explains the predictions of any classifier by approximating it locally with an interpretable model, helping users understand model behavior in specific instances.
- DiCE: dice_ml, a python library for counterfactual explanation

Backend Service / API:

- Node-JS: It retrieves the precomputed predictions and explanations from Firebase and serves them to the frontend.

Database:

- Firebase: we used firebase instead of MongoDB because it is more compatible with Google Colab. It was used for storing and retrieving the predictions and explanations in a JSON format.

3.1.2 Frontend (User Interaction and Visualization)**Programming Language:**

- JavaScript: Standard for interactive web applications.

Frameworks/Libraries:

- React.js: used to build the user interface

Survey Implementation:

- Qualtrics: used to build the survey due to its strong data analytics capabilities and support for randomization. The survey included S-TIAS and SCS questionnaires to evaluate user feedback on the explanations.

Visualization Tools:

- Custom React Components: used to display SHAP, LIME, and DiCE explanations

Styling:

- Tailwind CSS: For responsive and visually appealing designs.

3.1.3 Architecture Diagram:

The system architecture diagram:

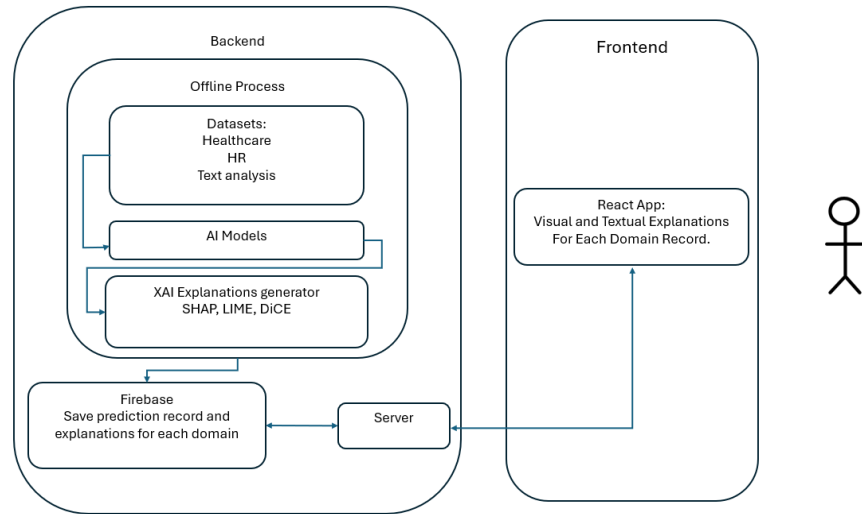


Figure 1 - System Architecture

Offline Phase (Left): Raw datasets from three domains (Healthcare, HR, Text) are processed by the AI models. The system then generates explanations using SHAP, LIME, and DiCE, storing the results in the database.

Online Phase (Right): The frontend retrieves the computed predictions and explanations via the server and presents them to the user.

3.2 GUI

The system prediction is presented in Figure 2. Users select a domain (HR, Healthcare, or System Analyst), view the input data, and receive the prediction generated by the AI model. The system also provides three explanation methods: SHAP, LIME, and DiCE, with example outputs shown in the Appendix A System GUI (see [Figure 3](#), [Figure 4](#), [Figure 5](#)).



Figure 2 - System GUI and Prediction and Input Data

3.3 Challenges

3.3.1 Explaining the Model's Decision

A key challenge in this project was explaining why the system produced a specific prediction. Machine learning models are complex, and even though SHAP, LIME, and DiCE provide explanations, their outputs are often technical and difficult to understand. For this reason, we decided to present the explanations in a more visual and simple way. This choice was made because the explanations are shown to users in questionnaires, so they need to be clear and easy to understand in order to get meaningful feedback.

3.3.2 Compatibility Issues with DiCE in Text Analysis

Another challenge we faced was using DiCE with the text analysis dataset. At first, the explanations were not meaningful because DiCE changed words in a way that did not make sense, and the text lost its meaning. To solve this, we added style-based features (such as writing characteristics) to the data and used them as inputs for the model, instead of relying only on the raw text. This helped DiCE generate more reasonable and understandable explanations. We made this decision in order to keep the explanations clear and meaningful for users.

3.3.3 Challenges in Building the Questionnaire (Qualtrics)

Another challenge we faced was building the questionnaire using Qualtrics. We had difficulties adding images, since they could not be uploaded directly and had to be inserted as links, which required us to convert the images into URLs and embed them using HTML. In addition, the images appeared too small, so we had to adjust our system and increase the text size to make the explanations more readable. We also found it challenging to implement randomization within the questionnaire, especially since it included three different datasets, which made the survey longer and more complex.

3.3.4 Getting Users to Participate in the Survey

Another challenge we faced was collecting responses from users. It was difficult to distribute the questionnaire and get enough participants. Since the survey was in English, many users found it hard to read and understand the explanations of the XAI tools, which affected their ability to respond. Consequently, we opted to outsource the survey to a dedicated platform to ensure a larger, more representative, and reliable data set.

4 Evaluation Process

To evaluate our system, we conducted a user study using a structured questionnaire built in Qualtrics. The survey was designed to assess how users perceive AI explanations across different domains with varying levels of risk: Healthcare, Human Resources (HR), and Text Analysis. Participants were first given a short introduction explaining the purpose of the study and the evaluation process. They were then asked to provide basic demographic information, such as gender and age, and to rank the different domains based on their perceived impact.

For each domain, participants were presented with a real example from our system, including the model’s prediction and only one explanation method (SHAP, LIME, or DiCE), shown as a visual output taken directly from our system. The order of the domains was randomized for each user, and for each domain, the explanation method was also randomly selected. This means that each participant saw all three domains, but received a different explanation type for each domain (for example, SHAP for HR, LIME for Healthcare, and DiCE for Text Analysis).

For each scenario, participants answered a set of Likert-scale questions (1 – Strongly Disagree to 5 – Strongly Agree), based on SCS (System Causability Scale) and S-TIAS, to evaluate the explanations. These questions focused on understanding, clarity, usability, and trust in the system’s decision.

The SCS score is calculated by assigning a score between 1 and 5 on a Likert scale to each of the 10 questions. The possible ratings are 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, and 5 = strongly agree. The ratings for the 10 categories are summed, and the resulting score is obtained by dividing the sum by 50. This yields scores ranging from 0.2 to 1.0, with 1.0 representing the optimal result [10]. Similarly, the S-TIAS score is calculated by summing the ratings of its three constituent items and dividing the total by 3 [23].

We chose the SCS framework because it measures whether an interface helps users understand the underlying causality of an AI decision [10]. In addition, we chose S-TIAS because it's a practical and valid alternative for measuring trust in automation and AI for the purposes of identifying antecedent factors of trust and predicting trust outcomes [23].

The full SCS and S-TIAS questionnaires are provided in the Appendix B (see [SCS questions](#), [S-TIAS questions](#)). The SCS framework includes 10 items, while the S-TIAS framework includes 3 items.

4.1 Testing Procedure

The testing procedure included validation of the explanation generation process, data storage, presentation of domains and explanations, user interaction, and response collection. A detailed description of all test cases is provided in the Appendix C (Table [C.1](#)).

4.2 Data Collection

The questionnaire was initially distributed through personal networks, including friends, family members, student groups within the college, and LinkedIn. However, the number of responses obtained was insufficient for the planned analysis. To increase participation, the survey was distributed through an outsourcing platform. This approach improved the response rate, resulting in a total of 104 completed questionnaires.

4.3 Data Cleaning and Response Filtering

To ensure data integrity, we excluded participants who failed the attention check, as well as those who completed the evaluation in under 2 minutes or over 15 minutes. Additionally, we discarded "straight-lining" responses where the same answer was selected for all items as these indicated a lack of attentive engagement.

Finished	Attention Check	Response time (2 min -15 min)	Same Answers (discard)
104	92	89	87

Table 2 - Response Filtering

5 Evaluation Results

5.1 Perceived Risk Study

Before evaluating the AI explanations, we asked participants to rank each domain based on how much it impacts their daily lives (on a scale of 1 to 10). The results show that participants perceive Finance (6.79) and Healthcare (6.61) as having the highest impact, while Text Analysis (4.97) and Human Resources (4.31) are perceived as having a lower impact.

5.2 Data Overview and Descriptive Statistics

As an initial step in the analysis, descriptive statistics were calculated for the SCS and the S-Tias metric across the three risk domains and the evaluated XAI methods (DICE, LIME, SHAP). The complete table detailing the means and standard deviations for each experimental condition is provided in Appendix D (see Table [D.1](#)).

The descriptive statistics show similar comprehension (SCS) and trust (S-Tias) scores across explanation methods in the Healthcare and HR domains. Specifically, SCS scores ranged closely from 0.643 to 0.671 in Healthcare, and from 0.608 to 0.655 in HR. A wider variation appeared in the Text Analysis (LLM) domain. Here, DiCE achieved the highest scores (SCS = 0.710, S-Tias = 0.57), while SHAP recorded the lowest (SCS = 0.569, S-Tias = 0.48).

5.3 Mixed Effect Model Results

A Mixed Effects Model was used to examine how risk domain, explanation method, and perceived risk affected the SCS and S-TIAS scores. Since each participant answered several questionnaires, this model was used to account for repeated responses from the same participant and to reduce the effect of individual differences. The reference group was defined as participants in the Healthcare domain who received a DiCE explanation. The full model results are presented in Appendix D (see [Figure 6](#) and [Figure 7](#)).

The main results are summarized in Table 3. Overall, the model showed that the Healthcare and HR domains did not differ significantly in either comprehension or trust scores. However, in the LLM domain, DiCE was associated with higher SCS and S-TIAS scores, while LIME and SHAP generally showed lower scores compared to DiCE. In addition, perceived risk had a small but significant positive effect on both measures, suggesting that participants who perceived the scenario as more important or risky tended to report slightly higher SCS and S-Tias scores.

Evaluation Measure	SCS Results	S-TIAS Results	Significance
Healthcare vs. HR Domain	–	–	No significant difference
LLM Domain with DiCE	Coef = 0.080 (p = 0.016)	Coef = 0.487 (p = 0.011)	Significant positive effect
LLM + LIME vs. DiCE	Coef = -0.101 (p = 0.042)	Coef = -0.557 (p = 0.053)	Negative effect; S-TIAS approached significance
LLM + SHAP vs. DiCE	Coef = -0.164 (p = 0.001)	Coef = -0.642 (p = 0.030)	Significant negative effect
Perceived Risk	Coef = 0.009 (p = 0.012)	Coef = 0.043 (p = 0.034)	Significant positive effect

Table 4- Summary of mixed effect model results

5.4 Tukey Post-Hoc Analysis

A Tukey Post-Hoc test was used to compare the explanation methods within each domain. The full results are presented in Appendix, [Figure 8](#), [Figure 9](#), and [Figure 10](#).

In the Healthcare and HR domains, no statistically significant differences were found between DiCE, LIME, and SHAP for either the SCS or S-TIAS scores. In the LLM domain, the results differed between the two metrics. For the SCS score, DiCE performed significantly better than SHAP, with a difference of 0.140 points (p = 0.005). The difference between DiCE and LIME was not statistically significant (p = 0.081), but the result approached statistical significance.

For the S-TIAS score, none of the pairwise comparisons reached statistical significance, which may be related to the higher variance in trust responses and the correction applied for multiple comparisons. However, the descriptive results showed lower trust scores for both LIME and SHAP compared to DiCE. LIME was lower by 0.500 points (p = 0.155), while SHAP was lower by 0.374 points (p = 0.368).

5.5 Friedman Test Analysis

A Friedman test was used to evaluate whether SCS and S-TIAS scores differed across the three risk domains while accounting for repeated measures from the same participants. The full results are presented in Appendix D [Figure 11](#) and [Figure 12](#).

The test showed no statistically significant differences between the domains for either metric. For the SCS score, no significant difference was found across the domains (Friedman statistic = 1.409, $p = 0.494$). Similarly, for the S-TIAS score, the differences between domains were not statistically significant (Friedman statistic = 4.222, $p = 0.121$). Therefore, based on this test, there was no statistical evidence that the domain alone affected participants' overall SCS or S-Tias scores.

5.6 Wilcoxon Post-Hoc Analysis

To follow up on the Friedman test, Wilcoxon signed-rank paired tests with Holm correction were used to perform direct pairwise comparisons between the three risk domains. The full results are presented in Appendix D, [Figure 13](#) and [Figure 14](#).

Consistent with the Friedman test, the pairwise comparisons showed no statistically significant differences between any pair of domains for either the SCS or S-TIAS scores. For the S-TIAS score, some unadjusted p-values initially suggested possible differences between domains. However, after applying the Holm correction for multiple comparisons, these differences were no longer statistically significant. Overall, these results provide no statistical evidence that the domain alone created a significant difference in participants' SCS or S-Tias scores.

5.7 Repeated Measures Correlation

A Repeated Measures Correlation was used to evaluate the relationships between perceived risk, comprehension (SCS), and trust (S-TIAS), while accounting for the multiple questionnaires completed by each participant. The full results are presented in Appendix D , [Figure 15](#).

A significant but weak positive correlation was found between perceived risk and the SCS score ($r = 0.163$, $p = 0.031$). In contrast, no significant correlation was found between perceived risk and the S-TIAS score ($r = 0.086$, $p = 0.258$). Finally, a strong and highly significant positive correlation was found between SCS and S-TIAS scores ($r = 0.64$, $p < 0.001$).

6 Discussion and Conclusions

This section summarizes the main conclusions drawn from the statistical analysis. Overall, the findings suggest that users' comprehension and trust were not shaped by the risk domain alone. Instead, the results point to a more context-dependent pattern, where the explanation method, the domain characteristics, and the user's perceived risk together influenced how XAI explanations were evaluated. The following sections discuss these conclusions in more detail.

6.1 The Role of the Explanation Method

The first main conclusion is that the explanation method influenced users' comprehension and trust, but this effect was not consistent across all domains. This conclusion is mainly supported by the Mixed Effects Model and the Tukey Post-Hoc test.

In the Healthcare and HR domains, no meaningful differences were found between DiCE, LIME, and SHAP. This suggests that in these domains, users evaluated the different explanation methods in a relatively similar way. However, the text-analysis domain showed a different pattern. The results indicated significant interaction effects between the text-analysis domain and the explanation method. In addition, the Tukey Post-Hoc test showed that DiCE performed significantly better than SHAP in terms of SCS. Therefore, DiCE appears to be more effective in the text-analysis domain, especially for supporting users' comprehension.

One possible explanation is that the text-analysis domain required less professional domain knowledge compared to Healthcare or HR, making it more accessible to participants. In this context, DiCE may have been easier to understand because it presents more direct counterfactual changes that show what could change the model's decision. In contrast, SHAP and LIME may have been less clear in the LLM domain because they rely on feature importance, which can be harder to interpret when the features are textual.

6.2 Descriptive Results Across Explanation Methods

The descriptive results showed that the average SCS and S-TIAS scores were relatively similar across the three domains, with no single domain clearly standing out as producing higher comprehension or trust overall. However, when examining the explanation methods within each domain, clearer differences appeared.

DiCE received the highest overall average scores, but its advantage was most visible in the LLM domain, where it also showed statistically stronger results for comprehension. This suggests that for text-based or LLM-related scenarios, DiCE may be a more suitable explanation method, as counterfactual explanations can provide users with a clearer and more direct understanding of what could change the model's decision.

In contrast, in the Healthcare and HR domains, no clear statistical advantage was found for one explanation method over another. Therefore, the results do not support a universal preference for a single XAI method across all domains. Instead, they suggest that the explanation method should be selected according to the domain context and the type of information being explained.

6.3 The Role of Domain and Perceived Risk

The domain itself did not significantly affect users' comprehension or trust. This is supported by the Friedman test and the Wilcoxon paired tests with Holm correction, which showed no statistically significant differences between the Healthcare, HR, and LLM domains for either SCS or S-TIAS.

While the domain selected to represent different risk levels did not create significant differences on its own, participants' perceived level of risk had a small positive effect on their evaluations. The Mixed Effects Model showed that higher perceived risk was associated with slightly higher SCS and S-TIAS scores. In addition, the Repeated Measures Correlation showed a weak but significant positive relationship between perceived risk and SCS. This suggests that when users perceived a scenario as riskier or more important, they may have paid more attention to the explanation and processed it more carefully, resulting in slightly higher comprehension scores.

However, this effect was stronger and more consistent for comprehension than for trust. The direct correlation between perceived risk and S-TIAS was not statistically significant, indicating that risk perception may play a more consistent role in shaping users' understanding of AI explanations than their trust in the system.

6.4 The Relationship Between SCS and S-Tias

The strong positive correlation between SCS and S-TIAS provides an important conclusion about how participants evaluated the explanations. Trust appeared to be strongly linked to users' comprehension of the explanation. Participants who reported that they understood the explanation also tended to report higher trust in the system.

This means that, in this study, a higher trust score was more meaningful when it was accompanied by higher understanding, because it suggests that users trusted the system after making sense of the explanation, rather than simply accepting the AI output. Therefore, one of the main conclusions is that understandable explanations are essential for meaningful trust in XAI systems. If users cannot understand the explanation, their trust in the system may be weaker.

6.5 Main Conclusions

The central question of this study was whether different XAI explanation methods affect users' comprehension and trust differently depending on the risk level of the domain. The findings provide a more complex answer: the risk level of the domain alone did not determine how users

responded to AI explanations. Instead, the results suggest that users' comprehension and trust were influenced mainly by the match between the explanation method and the domain context.

When the explanation method matched the nature of the domain, clearer differences appeared. In particular, DiCE performed better in the text-analysis domain, especially in terms of comprehension. This suggests that counterfactual explanations may be more suitable for text-based AI scenarios, where users can benefit from direct "what-if" examples that show what could change the model's decision. In contrast, in more structured domains such as Healthcare and HR, the choice of explanation method appeared less critical, since DiCE, LIME, and SHAP produced relatively similar results.

Perceived risk also played a role, but its effect was limited. Users who perceived a scenario as riskier tended to report slightly higher comprehension scores, suggesting that personal risk perception may increase users' attention or engagement with the explanation. However, perceived risk did not consistently increase trust, meaning that trust was not created simply because users viewed a scenario as more important or risky.

Finally, the strong relationship between comprehension and trust shows that understanding is central to how users evaluate XAI explanations. Users who understood the explanations better also tended to trust the system more. Therefore, trust in XAI should not be based only on providing detailed or technical explanations, but on providing explanations that users can actually understand.

Overall, the main conclusion is that the effectiveness of an XAI method is context-dependent. There is no single explanation method that should be assumed to work best across all domains. Instead, XAI explanations should be selected according to the domain, the type of data being explained, and the user's ability to understand the explanation.

6.6 Achievement of Project Objectives

Overall, the main objectives of the project were achieved. The study successfully examined how users respond to different XAI explanation methods across several domains and provided insight into the relationship between explanation methods, comprehension, trust, and perceived risk. Although the results did not identify one explanation method that clearly works best across all domains, they showed that the effectiveness of XAI explanations depends on the domain context, the type of information being explained, and how users perceive the scenario. Therefore, the project met its goal of evaluating XAI methods from the user's perspective and helped clarify the complexity of designing explanations that are both understandable and trustworthy.

6.7 Reflections on the Research Process

One of the main lessons learned from this project is the importance of making both the explanations and the questionnaire as clear and accessible as possible. Since participants were asked to evaluate AI explanations across different domains, overly complex wording or technical explanations may have made the task more difficult and may have affected participation. In future work, simplifying the explanations, using clearer questionnaire wording, and possibly adding short examples before the questions could help participants understand the task better and may lead to higher response rates and more reliable feedback.

Appendices

Appendix A – System GUI

Figure 3: SHAP Explanation In The System

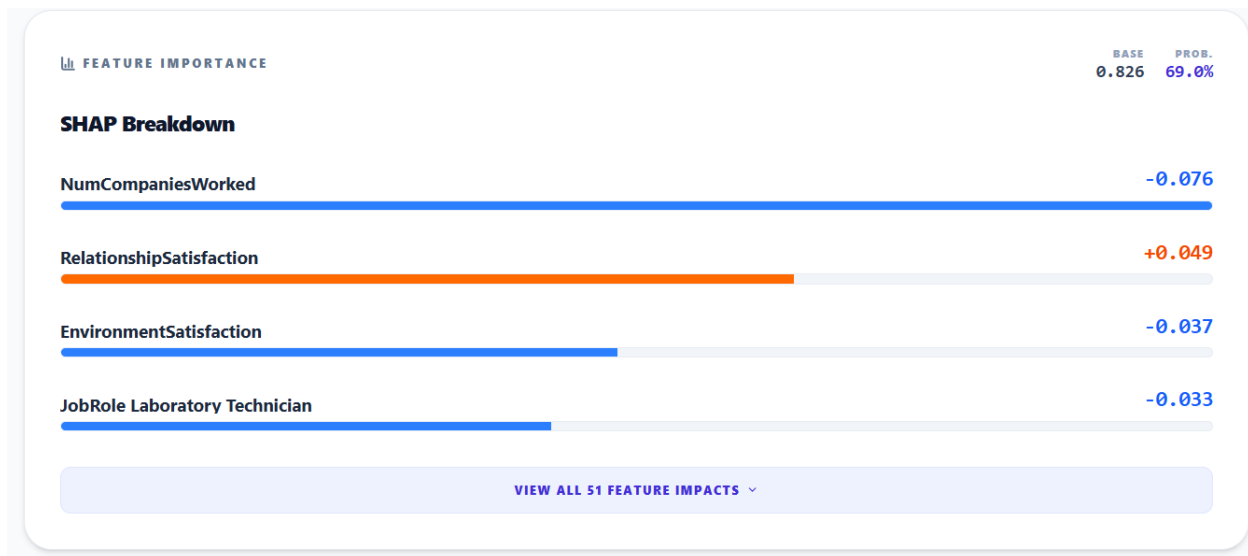


Figure 3 - SHAP output

Figure 4: LIME Explanation In The System

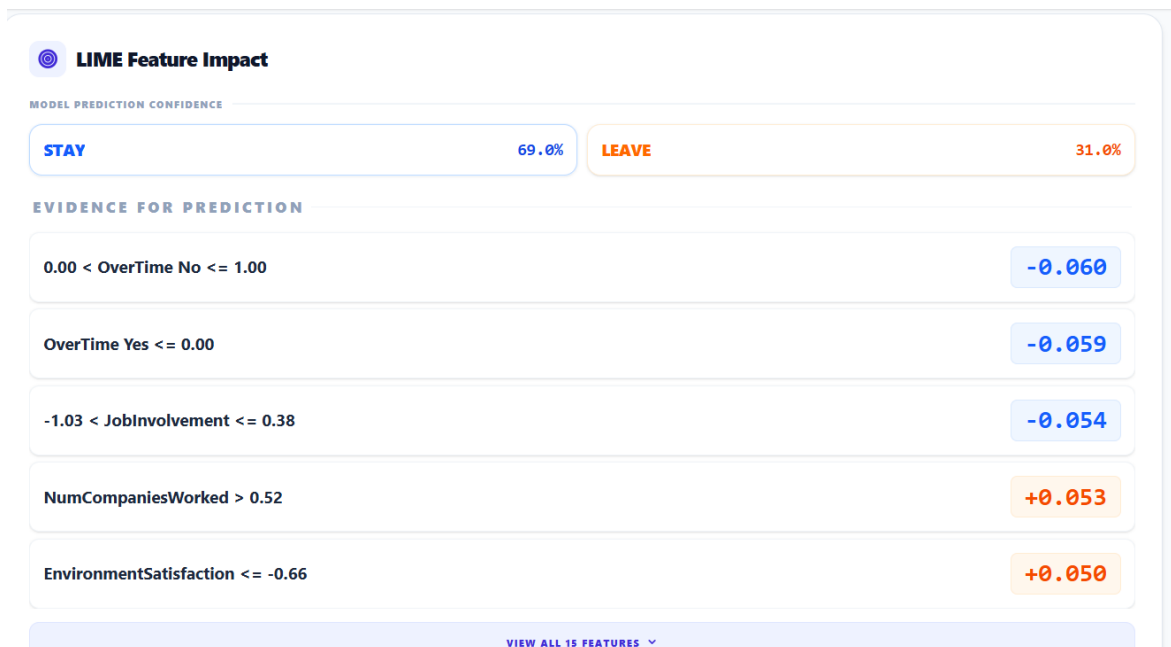


Figure 4 - LIME output

Figure 5: DiCE Explanation In The System



Figure 5 - DiCE output

Appendix B - Full SCS and S-TIAS Questionnaires

SCS questions

- I found that the data included all relevant known causal factors with sufficient precision and granularity.
- I understood the explanations within the context of my work.
- I could change the level of detail on demand.
- I did not need support to understand the explanations.
- I found the explanations helped me to understand causality.
- I was able to use the explanations with my knowledge base.
- I did not find inconsistencies between explanations.
- I think that most people would learn to understand the explanations very quickly.
- I did not need more references in the explanations (e.g., medical guidelines, regulations).
- I received the explanations in a timely and efficient manner.

S-Tias questions

- I found that I am confident in the system's results.
- I found that the system is reliable for making decisions.
- I found that I can trust the system's decision.

Appendix C - Testing Procedure

C.1: Testing Procedure Table

Test ID	Description	Expected Result	Precondition	Comments
1	Running Python explanation scripts	Python scripts generate SHAP, LIME, and DiCE explanations for each dataset.	Datasets and trained models are available.	One instance per domain.
2	Saving explanations to database	All generated explanations are saved in MongoDB.	Explanations are successfully generated.	Same explanations used for all users.
3	Presenting all domains	The system presents all three domains: Healthcare, HR, and Text Analysis.	Explanations are stored in the database.	Same domains shown to all users.
4	Loading predefined instances	The system loads one predefined decision instance for each domain.	Domains are presented.	Identical instances shown to all users for comparison.
5	Displaying SHAP explanations	The system displays SHAP explanations for each domain.	SHAP explanations are pre-generated.	Ensure clarity and consistent format.
6	Displaying LIME explanations	The system displays LIME explanations for each domain.	LIME explanations are pre-generated.	Same explanation structure across domains.
7	Displaying DiCE explanations	The system displays DiCE-based explanations for each domain.	DiCE explanations are pre-generated.	Explanations refer to the same decision instance.

8	User review of explanations	The user can review all explanations for each domain.	All explanations are displayed.	Explanations are presented without guidance.
9	Filling the evaluation form	The user fills in the evaluation Survey	Explanations have been reviewed.	Measures trust, clarity, and explainability.
10	Saving user responses	The Survey saves all user responses successfully.	Form is submitted.	Data stored for later analysis.

Table 1 - Testing Procedure Table

Appendix D - Statistical Results

D.1 Descriptive Statistics Table

Risk Domain	Explanation Method	Sample Size (N)	Mean SCS Score	Mean S-Tias Score	Mean S-Tias Normalized Score	Standard Deviation (SD) - SCS	Standard Deviation (SD) - S-Tias
Human Resources (HR)	SHAP	30	0.608	2.76666667	0.441666667	0.1667209107	1.043681959
	LIME	31	0.6148387097	2.720430108	0.4301075269	0.1619026656	0.8610793149
	DiCE	26	0.6553846154	2.974358974	0.4935897436	0.1649540729	0.9088360171
	Overall HR Domain	87	0.6245977011	2.812260536	0.4530651341	0.1638405677	0.9377736212
Healthcare (Diabetes)	SHAP	27	0.6718518519	3.12345679	0.5308641975	0.1746433076	0.9877374134
	LIME	26	0.6438461538	2.858974359	0.4647435897	0.1394869721	0.8494342008
	DiCE	34	0.6488235294	2.990196078	0.4975490196	0.161372235	0.8468014452
	Overall Healthcare Domain	87	0.6544827586	2.992337165	0.4980842912	0.1581488747	0.8936876471

Text Analysis (LLM)	SHAP	27	0.5696296 296	2.9259259 26	0.48148148 15	0.18054 36551	1.1410496 49
	LIME	30	0.618	2.8	0.45	0.17681 86291	1.1679209 1
	DiCE	30	0.71	3.3	0.575	0.13169 9763	1.0336227 88
	Overall LLM Domain	87	0.6347126 437	3.0114942 53	0.50287356 32	0.17219 40433	1.0497302 17

Table 3 - Descriptive Statistics Table

Figure 6: Mixed Effect Results For SCS

```

=====
.. SCS ANALYSIS - MIXED EFFECTS MODEL
=====

Mixed Linear Model Regression Results
=====
Model: MixedLM Dependent Variable: SCS_Score
No. Observations: 261 Method: ML
No. Groups: 87 Scale: 0.0135
Min. group size: 3 Log-Likelihood: 136.4556
Max. group size: 3 Converged: Yes
Mean group size: 3.0

-----
Coef. Std.Err. z P>|z| [0.025 0.975]
-----
Intercept 0.593 0.033 17.797 0.000 0.528 0.658
C(Domain)[T.HR] 0.031 0.034 0.914 0.361 -0.035 0.097
C(Domain)[T.LLM] 0.080 0.033 2.409 0.016 0.015 0.145
C(Explanation_Method)[T.LIME] 0.008 0.035 0.242 0.809 -0.060 0.077
C(Explanation_Method)[T.SHAP] 0.005 0.034 0.138 0.891 -0.063 0.072
C(Domain)[T.HR]:C(Explanation_Method)[T.LIME] -0.053 0.049 -1.078 0.281 -0.149 0.043
C(Domain)[T.LLM]:C(Explanation_Method)[T.LIME] -0.101 0.050 -2.034 0.042 -0.198 -0.004
C(Domain)[T.HR]:C(Explanation_Method)[T.SHAP] -0.065 0.049 -1.330 0.183 -0.160 0.031
C(Domain)[T.LLM]:C(Explanation_Method)[T.SHAP] -0.164 0.051 -3.220 0.001 -0.264 -0.064
Perceived_Risk 0.009 0.003 2.525 0.012 0.002 0.015
Group Var 0.011 0.026
=====

```

Figure 6 - Mixed Effects Model Analysis for SCS

Figure 7: Mixed Effect Results For S-TIAS

```
=====
STIAS ANALYSIS - MIXED EFFECTS MODEL
...
=====

Mixed Linear Model Regression Results
=====
Model: MixedLM Dependent Variable: STias_Score
No. Observations: 261 Method: ML
No. Groups: 87 Scale: 0.4499
Min. group size: 3 Log-Likelihood: -324.0053
Max. group size: 3 Converged: Yes
Mean group size: 3.0
-----

```

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	2.743	0.195	14.061	0.000	2.361	3.126
C(Domain)[T.HR]	0.100	0.194	0.516	0.606	-0.281	0.481
C(Domain)[T.LLM]	0.487	0.192	2.537	0.011	0.111	0.864
C(Explanation_Method)[T.LIME]	-0.136	0.202	-0.674	0.500	-0.533	0.260
C(Explanation_Method)[T.SHAP]	0.019	0.200	0.096	0.924	-0.373	0.411
C(Domain)[T.HR]:C(Explanation_Method)[T.LIME]	-0.162	0.285	-0.570	0.569	-0.720	0.396
C(Domain)[T.LLM]:C(Explanation_Method)[T.LIME]	-0.557	0.288	-1.935	0.053	-1.122	0.007
C(Domain)[T.HR]:C(Explanation_Method)[T.SHAP]	-0.338	0.281	-1.203	0.229	-0.889	0.213
C(Domain)[T.LLM]:C(Explanation_Method)[T.SHAP]	-0.642	0.296	-2.171	0.030	-1.222	-0.063
Perceived_Risk	0.043	0.020	2.125	0.034	0.003	0.083
Group Var	0.418	0.160				

```
=====
```

Figure 7 - Mixed Effects Model Analysis for S-Tias

Figure 8: Tukey Post Hoc Analysis for the HR Domain

```
-----
Domain: HR - SCS
-----

Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj lower upper reject
-----
DICE LIME -0.0405 0.6249 -0.1449 0.0638 False
DICE SHAP -0.0474 0.5321 -0.1525 0.0578 False
LIME SHAP -0.0068 0.9856 -0.1074 0.0937 False
-----

Domain: HR - STIAS
-----

Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj lower upper reject
-----
DICE LIME -0.2539 0.5701 -0.8514 0.3435 False
DICE SHAP -0.2077 0.6898 -0.8097 0.3943 False
LIME SHAP 0.0462 0.9799 -0.5292 0.6216 False
-----
```

Figure 8 - Tukey Post Hoc Analysis for the HR Domain

Figure 9: Tukey Post Hoc Analysis for the Diabetes Domain

```
-----
Domain: Diabeties - SCS
-----

Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj lower upper reject
-----
DICE LIME -0.005 0.9921 -0.1042 0.0942 False
DICE SHAP 0.023 0.8417 -0.0751 0.1212 False
LIME SHAP 0.028 0.7992 -0.0766 0.1326 False
-----

-----
Domain: Diabeties - STIAS
-----

Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj lower upper reject
-----
DICE LIME -0.1312 0.8397 -0.6867 0.4242 False
DICE SHAP 0.1333 0.8319 -0.4163 0.6828 False
LIME SHAP 0.2645 0.5308 -0.3213 0.8503 False
-----
```

Figure 9 - Tukey Post Hoc Analysis for the Diabetes Domain

Figure 10: Tukey Post Hoc Analysis for the LLM Domain

```
-----
Domain: LLM - SCS
-----
..
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj lower upper reject
-----
DICE LIME -0.092 0.0817 -0.193 0.009 False
DICE SHAP -0.1404 0.005 -0.2441 -0.0366 True
LIME SHAP -0.0484 0.5092 -0.1521 0.0554 False
-----

-----
Domain: LLM - STIAS
-----

Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj lower upper reject
-----
DICE LIME -0.5 0.1558 -1.1403 0.1403 False
DICE SHAP -0.3741 0.3682 -1.0319 0.2837 False
LIME SHAP 0.1259 0.8915 -0.5319 0.7837 False
-----
```

Figure 10 - Tukey Post Hoc Analysis for the LLM Domain

Figure 11: Friedman test for the LLM Domain

```
=====
FRIEDMAN TEST BETWEEN DOMAINS - SCS
=====
```

Domain	Diabeties	HR	LLM
Participant			
1	0.56	0.68	0.56
2	0.74	0.72	0.88
3	0.84	0.76	0.74
4	0.52	0.68	0.54
5	0.48	0.54	0.42

Friedman Statistic = 1.4092307692308057
p-value = 0.4942986525666401
Result: No significant difference between domains for SCS

Figure 11 - Friedman test for the LLM Domain

Figure 12 : Friedman test for the LLM Domain

```
=====
FRIEDMAN TEST BETWEEN DOMAINS - STIAS
=====
```

Domain	Diabeties	HR	LLM
Participant			
1	2.666667	3.666667	2.333333
2	3.000000	4.000000	4.666667
3	4.333333	4.000000	4.000000
4	2.000000	2.333333	2.000000
5	2.666667	1.666667	2.000000

Friedman Statistic = 4.222222222222845
p-value = 0.12110333239232593
Result: No significant difference between domains for STIAS

Figure 12 - Friedman test for the LLM Domain

Figure 13: Post Hoc Wilcoxon Pairwise Comparisons Between Domains (SCS)

```
=====
POST HOC BETWEEN DOMAINS - SCS
Wilcoxon Paired Tests with Holm Correction
=====
```

	Domain 1	Domain 2	Statistic	p-value	p-adjusted	Significant
0	Diabeties	HR	1232.5	0.062986	0.188958	False
1	Diabeties	LLM	1441.0	0.620096	1.000000	False
2	HR	LLM	1566.5	0.657924	1.000000	False

Figure 13 - Post Hoc Wilcoxon Pairwise Comparisons Between Domains (SCS)

Figure 14: Post Hoc Wilcoxon Pairwise Comparisons Between Domains (S-TIAS)

```
=====
POST HOC BETWEEN DOMAINS - STIAS
Wilcoxon Paired Tests with Holm Correction
=====
```

	Domain 1	Domain 2	Statistic	p-value	p-adjusted	Significant
0	Diabeties	HR	761.0	0.041061	0.118808	False
1	Diabeties	LLM	1021.0	0.588588	0.588588	False
2	HR	LLM	864.0	0.039603	0.118808	False

Figure 14 - Post Hoc Wilcoxon Pairwise Comparisons Between Domains (S-TIAS)

Figure 15: Repeated Measures Correlation: SCS and S-TIAS


```

-----
Repeated Measures Correlation: Perceived Risk vs SCS
-----
      r  dof    pval    CI95    power
rm_corr 0.162522 173 0.031646 [0.01, 0.3] 0.577675
Repeated Correlation Coefficient (r): 0.163
p-value: 0.03165
Result: Significant repeated measures correlation

-----
Repeated Measures Correlation: Perceived Risk vs STIAS
-----
      r  dof    pval    CI95    power
rm_corr 0.08581 173 0.258851 [-0.06, 0.23] 0.204448
Repeated Correlation Coefficient (r): 0.086
p-value: 0.25885
Result: No significant repeated measures correlation

-----
Repeated Measures Correlation: SCS vs STIAS
-----
      r  dof    pval    CI95    power
rm_corr 0.640052 173 1.487488e-21 [0.54, 0.72] 1.0
Repeated Correlation Coefficient (r): 0.640
p-value: 0.00000
Result: Significant repeated measures correlation

=====
Explanation Method Statistics
=====

```

Figure 15 - Repeated Measures Correlation: SCS and S-TIAS

Appendix E - System User Guide

System User Guide:

The goal of this platform is to show how AI models make predictions and how different explanation methods can help you understand those decisions.

Getting Started: The About Page

When you first enter, you will arrive at the About Page. This page gives you a full overview of the project and includes explanations of:

- Explainable AI (XAI): What it is and why it matters.
- The Domains: Background information on the three datasets used in this system (Healthcare, HR, and Text Analysis).
- The Explanation Methods: A breakdown of how SHAP, LIME, and DiCE work to explain AI decisions.

XAI Study USER HELP

ABOUT | HR | DIABETES | LLM

Explainable AI

This system presents our final project about **Explainable Artificial Intelligence (XAI)**. The goal of the project is to show how AI models make predictions and how explanation methods can help users understand those predictions.

Instead of showing only a final AI answer, the system displays the prediction together with explanations that describe which factors influenced the result. This makes the AI decision more transparent, understandable, and easier to trust.

What does the project include?

- Healthcare Analysis**
Evaluating patient health indicators to accurately detect diabetes.
- HR Analysis**
Understanding the drivers behind employee turnover and retention.
- Text Analysis**
Predicting if a text was human-authored or AI-generated.

How to use the system

- Select Domain**
Choose a dataset and a specific case to analyze.
- View Prediction**
See the AI's final answer and the input variables.
- Explore Logic**
Toggle between XAI methods to reveal the reasoning.

Explanation Methods

- SHAP**
The SHAP chart shows how much each specific feature contributes to the final prediction.
 - Features are ranked by total impact.
 - The length of the bar shows the influence.
- LIME**
LIME shows exactly which feature influenced this specific outcome the most. Each feature is given a score: some add weight pushing toward the final decision, while others remove weight from it.
- DiCE**
The DiCE table explains the AI's decision by providing "what-if" scenarios. It shows exactly what minimum changes to the data would be needed to flip the AI's decision to the opposite outcome.

Figure 16 - System's About Page

Navigating the System

You can move through the entire system using the top navigation bar.

- Domains: Select your area of interest directly from the top menu.
- User Help: This section provides clear explanations of SHAP, LIME, and DiCE.



Figure 17 - System's Navigation Bar

How to Use the System

1. Select a Domain

Use the top navigation bar to choose one of the following domains:

- Healthcare Analysis: Evaluates patient health indicators to accurately detect diabetes.
- HR Analysis: Understands the drivers behind employee turnover and retention.
- Text Analysis: Predicts if a text was human authored or AI generated.

2. View a Prediction

After picking a domain, select a specific case to analyze. The screen will display:

- The Input Variables: The data details for that specific case.
- The AI Prediction: The final answer the AI model generated.
- The 3 Different Explanations: DiCE, SHAP and LIME explanations for the prediction.

3. Review the explanations

The system explains why the AI made its choice. You can view the three different explanation methods to reveal the reasoning:

- SHAP: A chart that ranks features by their impact on the prediction.
- LIME: Highlights which specific features were most important for this individual prediction.
- DiCE: Shows alternative scenarios to help you understand what would need to change for the AI to make a different prediction.

AI Prediction:

STAY

INPUT DATA:

Overtime is **No**, Monthlyincome is **3468**, Joblevel is **1**, Department is **Research & Development**, Educationfield is **Medical**, and Age is **27**.

[VIEW ALL DATA](#)

Figure 2 - System GUI and Prediction and Input Data

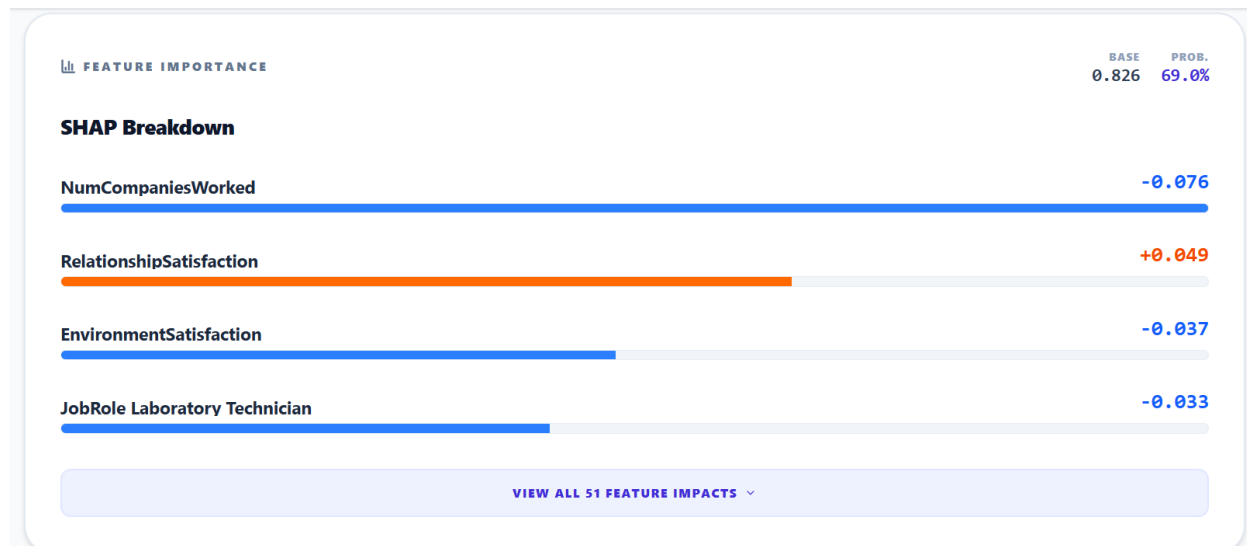


Figure 3 - SHAP output

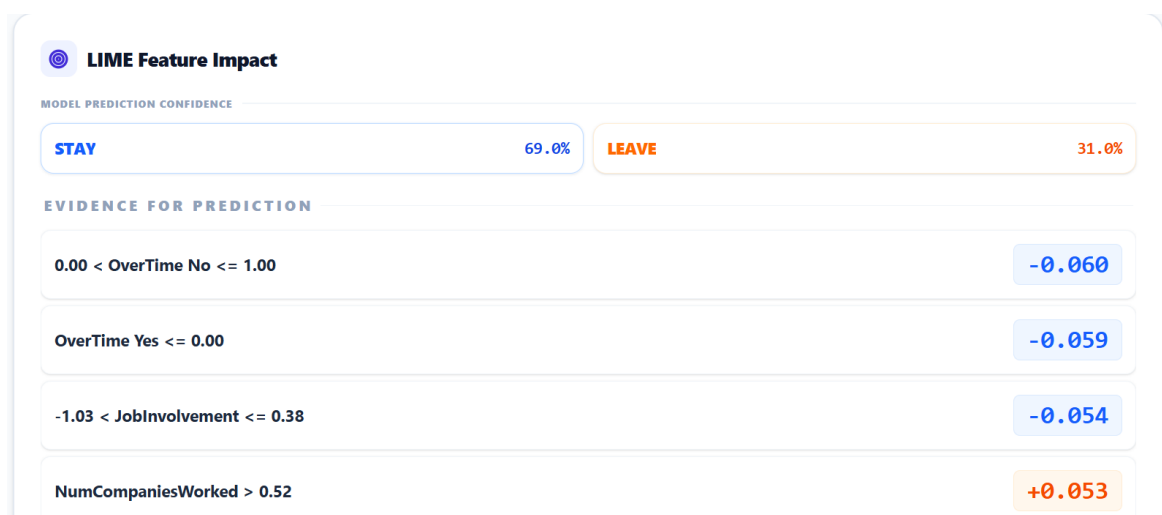


Figure 4 - LIME output

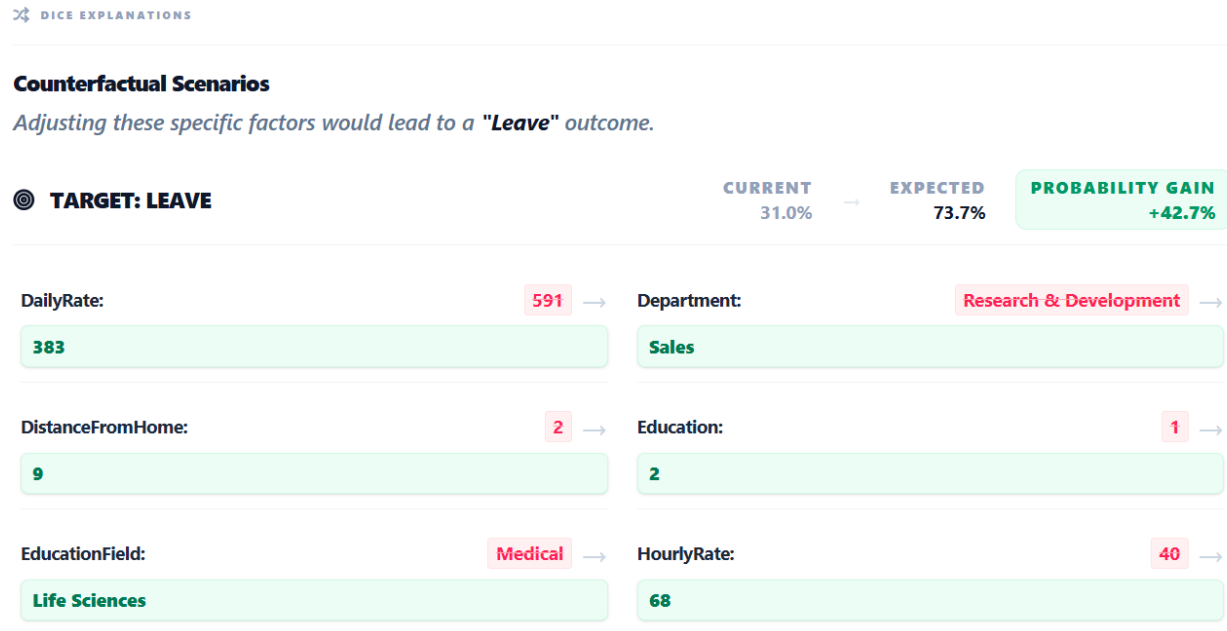


Figure 5 - DiCE output

Appendix F - Maintenance Guide

Maintenance Guide:

AI Models Management

The system utilizes two distinct machine learning models, developed and trained within the Google Colab environment using Python:

- **Healthcare Domain (DiabetesModel.ipynb):** A model trained on structured medical data to predict diabetes.
- **HR Domain (HRModel.ipynb):** A model trained on structured employee metrics to predict attrition.
- **Text Analysis Domain (IImModel.ipynb):** A model trained on textual datasets to distinguish between AI-generated and human-written content.

Models:

- **Logistic Regression:** Used for the **Text Analysis** domain. It is trained on textual vectors generated via TF-IDF Vectorization combined with stylistic feature extraction.
- **MLPClassifier (Multi-Layer Perceptron):** Used for the Healthcare and HR domains, trained on structured feature sets.

Data Processing and Pipeline Implementation

To ensure consistent model performance, all data undergoes a Pipeline workflow:

- **Preprocessing Techniques:** We utilize ColumnTransformer to apply preprocessing for each data type. Numerical features are normalized using StandardScaler and MaxAbsScaler (for stylistic features), while categorical features are processed via One-Hot Encoding.
- **Feature Integration:** In the text analysis domain, FeatureUnion is used to combine TF-IDF vectorization with custom stylistic feature extraction, ensuring a comprehensive feature set before model training.
- **Evaluation:** Model consistency is maintained using Stratified K-Fold Cross-Validation (k=5), ensuring robust performance across diverse data subsets.

Saving The Results and Deployment

The results (including the prediction value and the calculated SHAP, LIME, and DiCE explanations) **are saved directly to Firebase** as structured JSON objects. The web application retrieves these precomputed results for display.

JSON Schema Definition

The report object is structured as follows:

- **Prediction:** Contains the data for the prediction.
 - **record_id:** An identifier for the prediction instance.
 - **prediction:** The categorical output of the model.
 - **description:** A domain specific summary of the analysis context.
- **Original_Data:** Stores the raw input variables processed by the model.
- **explanations:** An object containing the precomputed interpretability data:
 - **lime:** Raw feature impact values generated by the LIME explainer.
 - **shap:** Feature contribution scores generated by the SHAP explainer.
 - **dice:** Counterfactual instances generated to show the minimal feature changes required to reach a different prediction outcome.

Usage in Frontend

The Web application retrieves this JSON object by its **record_id**. The frontend dynamically maps the explanations object to the corresponding components.

Web System Architecture

The system is built as a web based application, structured into a React.js frontend and a Node.js/Express.js backend (API). The deployment is fully managed via Vercel.

- **Frontend (User Interface):** Developed using React.js with Tailwind CSS for responsive, modern design. The interface is designed to dynamically render predictions and XAI explanations based on JSON objects retrieved from the database.
- **Backend (API Layer):** A Node.js/Express.js service acts as the communication bridge. It handles secure requests to the Firebase (Firestore) database, ensuring that precomputed prediction reports are correctly queried and served to the frontend.
- **Dynamic Visualization:** Components are built to interpret the JSON structure described previously, mapping explanation data (SHAP, LIME, DiCE) to their respective visualization components.
- **Deployment:** The application is deployed on Vercel. Every update pushed to the GitHub repository automatically triggers a re-deployment,

Current Constraints

Currently, the system is configured to retrieve and display a single primary prediction per domain from the database.

Future Improvements

- **Prediction Selector:** Adding a toggle or dropdown menu that allows users to browse and select between multiple historical predictions stored in the database.
- **Enhanced Visualizations:** Refining the UI design and layout of the SHAP, LIME, and DiCE to make the explanations more intuitive and visually engaging.

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